



Numerical analysis of added mass and damping of elastic hydrofoils *

Jia-sheng Li^{1,2,3}, Ye-gao Qu³, Hong-xing Hua³

1. School of Naval Architecture and Ocean Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

2. Hubei Provincial Engineering Research Center of Data Techniques and Supporting Software for Ships (DTSSS), Wuhan 430074, China

3. State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University, Shanghai 200240, China

(Received March 22, 2017, Revised August 31, 2020, Accepted September 1, 2020, Published online October 29, 2020)

©China Ship Scientific Research Center 2020

Abstract: A numerical model is proposed for analyzing the effects of added mass and damping on the dynamic behaviors of hydrofoils. Strongly coupled fluid-structure interactions (FSIs) of hydrofoils are analyzed by using the 3-D panel method for the fluid and the finite element method for the hydrofoils. The added mass and damping matrices due to the external fluid of the hydrofoil are asymmetric and computationally inefficient. The computational inefficiencies associated with these asymmetric matrices are overcome by using a modal reduction technique, in which the first several wet mode vectors of the hydrofoil are employed in the analysis of the FSI problem. The discretized system of equations of motion for the hydrofoil are solved using the Wilson- θ method. The present methods are validated by comparing the computed results with those obtained from the finite element analysis. It is found that the stationary flow is sufficient for determining the wet modes of the hydrofoil under the condition of single-phase potential flow and without phase change. In the case of relatively large inflow velocity, the added damping of the fluid can significantly affect the structural responses of the hydrofoil.

Key words: Fluid-structure interaction (FSI), hydroelastic response, hydrofoil, added mass, added damping

Introduction

A deep insight into the physical mechanism of fluid-structure interaction (FSI) is of great importance for the design of many hydropower structures, such as marine propellers and hydrofoils. In most cases, the strongly coupled FSI problem must be taken into account in the analysis and design of these structures. A key issue related to FSI analyses of underwater structures is to take into account the influences of the added mass and damping of the fluid, which are expected to have significant effects on the natural frequencies, mode shapes and structural responses of underwater structures. In addition, potential instability limits related to the resonances and flutter of underwater structures are also dependent upon the

added mass and damping of the fluid.

Much of the earlier research effort on FSI problems has been devoted to aerospace and aeroplane structures. For these FSI problems, the density of the fluid is much smaller than that of the structure, and therefore, the effects of damping and inertia forces of the fluid on the structural responses of the structure may be neglected. Nevertheless, the FSI problems of underwater structures involve high-density fluid, i.e. water, and in general, the effects of inertia and damping forces of the fluid on the structural responses of the structures may not be ignored. In recent years, a number of numerical methods have been proposed to study the effects of added mass and damping on the vibrations of hydrofoils. Seeley et al.^[1] designed three hydrofoils to investigate the added damping due to the interaction between the fluid and the hydrofoils by using piezocomposite actuators. The natural frequencies and damping were obtained based on experimental results. It was found that the natural frequencies of the hydrofoils were not substantially affected by the fluid, but the damping ratios were observed to increase in a linear manner with respect to the velocity of the fluid. Monette et al.^[2] developed a

* Project supported by the National Natural Science Foundation of China (Grant Nos. 52001130, 11922208 and 51839005).

Biography: Jia-sheng Li (1987-), Male, Ph. D., Assistant Professor, E-mail: lijiaosheng@hust.edu.cn

Corresponding author: Hong-xing Hua, E-mail: hxx@sjtu.edu.cn

mathematical model for studying the relationship between the hydro-dynamic damping and the flow velocity, and the hydro-damping of cantilever beams and runner blades were examined. They found that the hydro-dynamic damping increases linearly with the flow velocity, which was also observed by Seeley et al.^[1] La Torre et al.^[3] experimentally investigated the influence of the cavitation and supercavitation sheets of the leading edge on the added mass of a 2-D NACA0009 truncated hydrofoil by applying a non-intrusive excitation and measuring system. The results indicated that the maximum added mass effect can be achieved when cavitations appear, the added mass decreases as the cavity length is increased, and the added mass achieves minimum in the case of super cavitation. Kramer et al.^[4] examined the free vibrations of cantilevered composite plates immersed in air and water using combined analytical and numerical methods. The results revealed that the natural frequencies of the plate in water are about 50%-70% lower than those of the plate in air due to the significant effects of added mass. In addition, the added mass was found to vary considerably with the material orientation of the plate due to the bending-twist coupling of the anisotropic material. Chae et al.^[5] developed a combined numerical and experimental model for predicting the flow-induced vibrations of flexible hydrofoils. The natural frequencies and fluid damping coefficients were found to vary with the velocity, the angle of attack, and the added mass ratio of the solid to the fluid. Moreover, the ratios of natural frequencies for the hydrofoil immersed in water to those of hydrofoil in air decrease rapidly when the added mass and damping coefficients are increased rapidly. Lelong et al.^[6] studied the hydroelastic responses of flexible light weight hydrofoils under various flow conditions, including the case of unsteady partial cavitating flow in a hydrodynamic tunnel. The static deformation, strains, stresses and the vibrations of a rectangular cantilevered flexible hydrofoil made of polyoxymethylene plastic material were obtained. It was observed that the frequencies related to the bending vibration modes change slightly depending on the oscillation frequency of the cavity. The frequencies of the twisting modes tend to increase with respect to the cavity length, which may be related to the decrease of the added mass effects in the presence of vapor cavity on the surface of the hydrofoil. Liaghat et al.^[7] proposed a 3-D two-way FSI model to investigate the coupled effects of flowing fluid on a simplified hydrofoil. They found that the FSI must be included in the modeling in order to analyze the influence of the fluid on the vibration behaviors of the hydrofoil. However, strongly coupled FSI might be ignored for the prediction of the frequencies of the fluctuating

fluid forces if the main concern of the analysis is to check the possibility of resonance. Liu et al.^[8] used a coupled acoustic structural finite element method to study the FSI of a hydrofoil in cavitating flow. The results show that the added mass effect of the fluid varies against with the values of the cavitation surface ratio and with the thickness of the cavitation sheet. Cao et al.^[9] analyzed the hydrodynamic behaviors of an elastic hydrofoil. The results revealed that the cavitation number and angle of attack can significantly affect the hydrodynamic responses of the hydrofoil. Wu et al.^[10] investigated the hydroelastic response of a flexible NACA66 hydrofoil in cavitating flows based on experiments and numerical methods. Significant interactions were observed between the cavitation development and the hydroelastic response of the hydrofoil, and the hydrodynamic load coefficients of the flexible hydrofoil fluctuate more significantly than those of the rigid hydrofoil due to flow-induced flutter and deformation of the flexible hydrofoil. Astolfi et al.^[11] analyzed the hydro-elastic responses of flexible hydrofoils for various flow conditions including the cavitating flow. It was observed that increases of the flow velocity and the angle of attack do not change the bending mode frequencies of the hydrofoil, but they will increase the twisting mode frequency of the hydrofoil. In addition, the cavitation condition also changes the added-mass of the flexible hydrofoil. Hu et al.^[12] employed the finite volume method and the finite element method for studying the hydro-elastic problem of a 3-D cantilevered hydrofoil in the water tunnel. They found that small amplitude vibration of the hydrofoil has little effect on the development of the partial cavity and frequency spectrum of the lift containing the first bend frequency of the hydrofoil. Smith et al.^[13] investigated the cloud cavitation behavior around a hydrofoil due to the effect of FSI, and hydrofoil compliance was seen to dampen the higher frequency force fluctuations while showing strong correlation between normal force and tip deflection. Zeng et al.^[14] performed unsteady CFD and two-way FSI numerical simulation methods to investigate the effect of the trailing edge shape on added mass and hydrodynamic damping for NACA 0009 hydrofoils.

The objective of the present paper is to develop a highly efficient numerical method to analyze the added mass and damping matrices of an elastic hydrofoil moving in water. In the method, the kinematic boundary conditions imposed on the hydrofoil surface for the FSI analysis are derived based on the non-penetration conditions. The finite element method is applied to formulate the structural model of the hydrofoil, and a frequency-dependent panel method is used to derive the added mass and damping matrices due to the fluid. A modal reduction

technique combined with the Wilson- θ method is employed to calculate the structural responses of the hydrofoil. This overcomes the low computational efficiency of the analysis due to the asymmetric added matrices of the fluid. The results obtained by the present method are compared with those solutions obtained from the finite element analysis, and good agreement is achieved. The effects of the added mass and damping on the structural responses of the hydrofoil are investigated.

1. Formulation

1.1 Governing equations of the hydrofoil

We consider a 3-D hydrofoil immersed in a uniform flow, and the Cartesian $o-xyz$ coordinate system is employed to analyze the added mass and damping of the hydrofoil; see Fig. 1. The hydrofoil is constructed by a homogeneous and isotropic material. Since the geometrical configuration of the hydrofoil is complex, the finite element method is employed to obtain the discretized equations of motion for the structure. A linear isoparametric and eight-noded element with a total of 24 degrees of freedom is adopted in the finite element analysis for the hydrofoil. Using Lagrange's equation, the discretized equations of motion for the elastic hydrofoil is obtained as

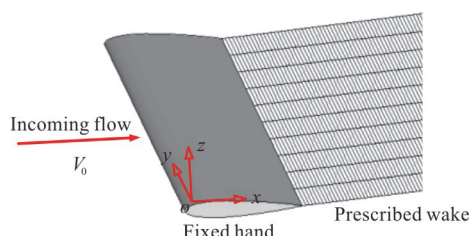


Fig. 1 (Color online) The geometrical configuration and coordinate system of the hydrofoil

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{F}_s + \mathbf{F}_f \quad (1)$$

where $\ddot{\mathbf{q}}$, $\dot{\mathbf{q}}$ and $\mathbf{q}$ are the nodal acceleration, velocity, and displacement vectors, respectively. $\mathbf{M}$ and $\mathbf{K}$ are the global mass and stiffness matrices assembled by those matrices of each element, given as: $\mathbf{M}^e = \int_{V_e} \rho_s \mathbf{N}^T \mathbf{N} dV$, $\mathbf{K}^e = \int_{V_e} \mathbf{B}^T \mathbf{D} \mathbf{B} dV$, where

$\mathbf{M}^e$ and $\mathbf{K}^e$ are the mass and stiffness matrices of the e^{th} element. $\mathbf{N}$, $\mathbf{B}$ and $\mathbf{D}$ are the shape function matrix, strain-displacement matrix, and material constitutive matrix of each element, respectively. The Rayleigh damping model is employed here to consider the damping effect of the hydrofoil. In doing so, the damping matrix of the

hydrofoil can be obtained as: $\mathbf{C} = \alpha \mathbf{M} + \beta \mathbf{K}$, where α and β are the two preassigned constants. $\mathbf{F}_s$ is the external force vector due to the structural forces applied on the hydrofoil. $\mathbf{F}_f$ is the hydrodynamic force vector.

1.2 Governing equations of the fluid

To determine the hydrodynamic force of the hydrofoils, some governing equations for the flowing fluid are applied, such as Navier-Stokes equations^[15-17] for viscous fluid, Euler's equations^[18] for rotational fluid and Laplace equations^[19] for inviscid and irrotational fluid. Although solving the first two types of equations may give a more accurate solution, they need lots of CPU time and memory. The solving of Laplace equations overcomes these disadvantages and is widely used for the analysis of hydrofoils. Hence, the Laplace equations and its commonly solving algorithms, panel method, are applied to calculate the hydrodynamic force vector $\mathbf{F}_f$. The fluid is assumed to be incompressible, inviscid and irrotational. The total velocity $\mathbf{V}_{\text{total}}$ of the fluid can be expressed in the sum of the uniform inflow velocity $\mathbf{V}_0$ and the disturbed fluid velocity due to a perturbation potential, given as

$$\mathbf{V}_{\text{total}} = \mathbf{V}_0 + \nabla \phi \quad (2)$$

where ϕ is the perturbation velocity potential corresponding to the flow field induced by the hydrofoil. Based on the potential flow theory and the kinematic boundary conditions on the hydrofoil surface, the hydrofoil-induced perturbation potential ϕ can be written as:

$$\nabla^2 \phi = 0 \quad \text{in the fluid domain } \Omega \quad (3)$$

$$-\frac{\partial \phi}{\partial \mathbf{n}} = \mathbf{V}_0 \cdot \mathbf{n} - \frac{\partial \delta}{\partial t} \cdot \mathbf{n} \quad \text{on hydrofoil surface } \Gamma \quad (4)$$

$$\Delta \phi_w(\mathbf{R}_{\text{wake}}, t) = \Delta \phi(\mathbf{R}_{re}, t - t') \quad \text{Morino Kutta condition} \quad (5)$$

where $\mathbf{n}$ is the outward unit normal vector. $\partial \phi / \partial \mathbf{n}$ represents the derivative of ϕ with respect to the normal direction. δ is the displacement vector of the nodal points on the hydrofoil surface. $\Delta \phi_w(\mathbf{R}_{\text{wake}}, t)$ represents the potential jump across the wake sheets. $\Delta \phi(\mathbf{R}_{re}, t - t')$ denotes the potential jump across the

hydrofoil surface at the trailing edge, which is equal to the potential at the upper side (suction side) minus that at the lower side (pressure side). t' is the time required for the fluid to travel along the wake surface from the hydrofoil trailing edge $\mathbf{R}_{re}$ to the wake point $\mathbf{R}_{wake}$. The kinematic boundary condition defined in Eq. (4) implies that the flow can slide in the tangent direction on the surface of the oscillating hydrofoil. Mathematically, this results in a Neumann-type boundary condition. Regarding the details of the Morino's Kutta condition defined in Eq. (5), the reader is referred to Morino and Kuo^[20].

Since linear dynamic problems are considered in this work, the hydrofoil-induced perturbation potential ϕ can be expressed as the sum of two terms, given as: $\phi = \phi_s + \phi_v$, for detailed proof, the reader is referred to Appendix A.

In general, the hydrodynamic forces due to the uniform flow V_0 (or the perturbation potential ϕ_s as a rigid hydrofoil advancing in the uniform flow) are not important in the dynamic analysis of the hydrofoil. Therefore, the hydrodynamic force vector depends on the perturbation potential ϕ_v , given as

$$\mathbf{F}_f = \mathbf{F}_p \quad (6)$$

where $\mathbf{F}_p$ represents the hydrodynamic force vector due to the perturbation potential, resulting in the added mass and damping matrices of the fluid. In order to improve the computational efficiency, a frequency-dependent panel method is employed to determine the hydrodynamic pressure vector $\mathbf{F}_p$ due to structural deformation of the elastic hydrofoil.

The governing equations for the perturbation potential due to the vibration of the flexible hydrofoil in a uniform flow can be written as:

$$\nabla^2 \phi_v = 0 \quad \text{in fluid domain } \Omega \quad (7)$$

$$\frac{\partial \phi_v}{\partial \mathbf{n}} = \frac{\partial \delta}{\partial t} \cdot \mathbf{n} \quad \text{on hydrofoil surface } \Gamma \quad (8)$$

$$\Delta \phi_{w,v}(\mathbf{R}_{wake}, t) = \Delta \phi_v(\mathbf{R}_{re}, t - t') \quad \text{Morino Kutta condition} \quad (9)$$

According to Eq. (7), the perturbation velocity potential ϕ_v is governed by Laplace's equation, and ϕ_v can be calculated by the following integral expression

$$2\pi\phi_v(\mathbf{R}_p, t) = \int_{S_h} \phi_v(\mathbf{R}_Q, t) \frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) dS -$$

$$\int_{S_h} G(\mathbf{R}_p, \mathbf{R}_Q) \frac{\partial}{\partial n_Q} \phi_v(\mathbf{R}_Q, t) dS + \int_{S_w} \Delta \phi_{w,v}(\mathbf{R}_Q, t) \frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) dS \quad (10)$$

where S_h and S_w represent the hydrofoil surface and wake sheet surface, respectively. Prescribed wake model is applied, and the angle of wake sheet is assumed to be average value of the attack angle and arch angle at the trailing edge. The subscript Q represents the variable point in the integration, and the subscript P represents the control point on the panel face. $\mathbf{R}_p$ and $\mathbf{R}_Q$ are the position vectors of the points on the hydrofoil face and wake sheet surface, respectively. $G(\mathbf{R}_p, \mathbf{R}_Q)$ is the fundamental solution (known as Green's function) of the Laplace equation in an unbounded 3-D fluid domain, given by: $G(\mathbf{R}_p, \mathbf{R}_Q) = 1/|\mathbf{R}_p - \mathbf{R}_Q|$. $\mathbf{n}_Q$ is the outward unit normal vector at point Q . $\Delta \phi_{w,v}(\mathbf{R}_Q, t)$ represents the potential jump across the wake sheets.

Substituting Eq. (8) into Eq. (10), one obtains

$$2\pi\phi_v(\mathbf{R}_p, t) - \int_{S_h} \phi_v(\mathbf{R}_Q, t) \frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) dS - \int_{S_w} \Delta \phi_{w,v}(\mathbf{R}_Q, t) \frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) dS = \int_{S_h} \left[-\frac{\partial}{\partial t} \delta(\mathbf{R}_Q, t) \cdot \mathbf{n}_Q \right] G(\mathbf{R}_p, \mathbf{R}_Q) dS \quad (11)$$

Assuming that the frequency of the excitation force for the linear dynamic system is k , and $\delta(\mathbf{R}_Q, t) = \Re[\delta_k(\mathbf{R}_Q)e^{-ikt}]$ is suitable, where $\Re$ means the real part of the expression, then $\phi_v(\mathbf{R}_Q, t)$ can be expressed as: $\phi_v(\mathbf{R}_Q, t) = \Re[\phi_{k,v}(\mathbf{R}_Q)e^{-ikt}]$. In order to solve Eq. (11), the surface of the hydrofoil is divided into N_s sub-panels in the spanwise direction, and in the chordwise direction, the surface is decomposed into N_c sub-panels from the leading edge to the trailing edge. The strengths of the dipoles and sources are assumed to be uniform on the surface of the panel. The wake field ranging from the trailing edge to sufficient distance downstream is divided into N_w sub-intervals. The discretized form of Eq. (11) can be written as

$$\{\phi_{k,v}\} = ikU\{\delta_{kn}\} \quad (12)$$

where U is the panel coefficient matrix, and

$$\{\delta_{kn}\} = [\delta_k(\mathbf{R}_1) \cdot \mathbf{n}(\mathbf{R}_1), \delta_k(\mathbf{R}_2) \cdot \mathbf{n}(\mathbf{R}_2), \dots, \delta_k(\mathbf{R}_N) \cdot \mathbf{n}(\mathbf{R}_N)]^T,$$

$$\{\phi_{k,v}\} = [\phi_{k,v}(\mathbf{R}_1), \phi_{k,v}(\mathbf{R}_2), \dots, \phi_{k,v}(\mathbf{R}_N)]^T$$

in which N is the total number of panels.

Applying the momentum equation of the fluid, one obtains Bernoulli's equation for potential flow as

$$p = p_0 + \rho \left(\frac{1}{2} V_0^2 - \frac{1}{2} V_{\text{total}}^2 - \frac{\partial \phi_v}{\partial t} \right) \quad (13)$$

where p_0 is the hydrostatic pressure at the computational point, ρ is the density of the fluid. In the present model, the absolute hydrostatic pressure is not important, and higher-order terms in Eq. (13) can be neglected if the vibration-induced perturbation velocity $\nabla \phi_v$ is small. In such cases, Eq. (13) may be rewritten as

$$p = -\rho \left(\frac{\partial \phi_v}{\partial t} + \mathbf{V}_0 \cdot \nabla \phi_v \right) \quad (14)$$

The above equation is applied to determine the pressure due to small deformations. There are two ways to calculate the unsteady pressure based on the Eq. (14), namely either using differential operation on the hydrofoil surface to calculate $\nabla \phi_v$ or using a direct integration method which requires robust numerical treatment of six singular integrals^[21-25]. The direct integration method is employed herein, and $\nabla \phi_v$ can be calculated by taking the i th component of the spatial derivative of Eq. (11) as follows

$$\begin{aligned} 2\pi \frac{\partial \phi_v(\mathbf{R}_p, t)}{\partial x_i} - \iint_{S_h} \phi_v(\mathbf{R}_Q, t) \frac{\partial}{\partial x_i} \left[\frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) \right] dS - \\ \iint_{S_h} \Delta \phi_{w,v}(\mathbf{R}_Q, t) \frac{\partial}{\partial x_i} \left[\frac{\partial}{\partial n_Q} G(\mathbf{R}_p, \mathbf{R}_Q) \right] dS = \\ \iint_{S_h} \left[-\frac{\partial \delta(\mathbf{R}_Q, t)}{\partial t} \cdot \mathbf{n}_Q \right] \frac{\partial}{\partial x_i} [G(\mathbf{R}_p, \mathbf{R}_Q)] dS \quad (15) \end{aligned}$$

Substituting Eqs. (9), (12) into Eq. (15), the discretized form of Eq. (15) can be written as

$$\left\{ \frac{\partial \phi_{k,v}}{\partial x_i} \right\} = ik[U]_{x_i} \{\delta_{kn}\} \quad (16)$$

where $[U]_{x_i}$ is the panel coefficient matrix, and

$$\left\{ \frac{\partial \phi_{k,v}}{\partial x_i} \right\} = \left[\frac{\partial \phi_{k,v}}{\partial x_i}(\mathbf{R}_1), \dots, \frac{\partial \phi_{k,v}}{\partial x_i}(\mathbf{R}_N) \right]^T$$

Substituting Eqs. (12), (16) into Eq. (14), one obtains the dynamic pressure as

$$\mathbf{p}_v(k) = \mathcal{R} \left[-\rho e^{-ikt} \left(ik \sum_{l=1}^3 [V_0]_{x_l} [U]_{x_l} \delta_{kn} + k^2 U \delta_{kn} \right) \right] \quad (17)$$

where $[V_0]_{x_i}$ is defined as: $[V_0]_{x_i} = \text{diag}\{V_{0x_i}(\mathbf{R}_1), \dots, [V_0]_{x_i} = \text{diag}\{V_{0x_i}(\mathbf{R}_1), \dots, V_{0x_i}(\mathbf{R}_N)\}$, and $V_{0x_i}(\mathbf{R}_i)$ is the component of $\mathbf{V}_0$ in the x_i direction on point $\mathbf{R}_i$. The pressure vector $\mathbf{p}_v(k)$ in Eq. (17) is expressed in terms of the unknown displacement vector δ_{kn} , which is calculated at hydrofoil panel centroids.

The fluid pressure vector may be expressed in terms of the nodal displacements of the finite element model of the hydrofoil. For the sake of brevity, the mesh of finite element model is assumed to be coincident with that of panel model on the surface of the hydrofoil. Assuming $\delta_{kn} = \mathbf{Z} \mathbf{q}_k$, where $\mathbf{Z}$ is shape function matrix and $\mathbf{q}_k$ is the nodal displacement vector, the dynamic force is obtained as

$$\mathbf{F}_v(k) = \mathcal{R} \left[\rho \mathbf{Z}^T \Delta s \left(ik \sum_{l=1}^3 [V_0]_{x_l} [U]_{x_l} + k^2 U \right) \mathbf{Z} \mathbf{q}_k e^{-ikt} \right] \quad (18)$$

where Δs is a diagonal matrix, expressed as: $\Delta s = \text{diag}\{\Delta s_1, \Delta s_2, \dots, \Delta s_N\}$. Δs_i is the area of the i th panel.

To calculate the perturbation potential $\phi_{k,v}$ and the hydroelastic force $\mathbf{F}_v(k)$ via the panel method, the nodal displacement vector $\mathbf{q}_k$ of the hydrofoil is required. In turn, the finite element analysis for the structural responses (displacement $\mathbf{q}_k$, velocity $\dot{\mathbf{q}}_k$ and acceleration $\ddot{\mathbf{q}}_k$) of the hydrofoil requires the hydrodynamic force $\mathbf{F}_v(k)$ from the fluid model based on the panel method. This is a typical FSI problem. The governing equations of the fluid and hydrofoil may be solved by placing the hydroelastic forces to the left-hand side of the

equilibrium equation of motion of the hydrofoil, given as

$$[M + M_w(k)]\ddot{q}_k + [C + C_w(k)]\dot{q}_k + Kq_k = F \quad (19)$$

where $M_w(k)$ and $C_w(k)$ are the added mass matrix and added damping matrix respectively. The detailed expressions of these two matrices are given by:

$$M_w(k) = \rho Z^T \Delta s [\mathcal{H}(U)] Z - \frac{1}{k} \rho Z^T \Delta s \sum_{l=1}^3 [V_0]_{x_l} \mathcal{I}([U]_{x_l}) Z \quad (20)$$

$$C_w(k) = \rho k Z^T \Delta s [\mathcal{I}([U])] Z + \rho Z^T \Delta s \sum_{l=1}^3 [V_0]_{x_l} \mathcal{H}([U]_{x_l}) Z \quad (21)$$

where $\mathcal{I}$ represents the imaginary part of an expression.

1.3 Solution scheme for the governing equations

Direct time-integration methods, such as the Wilson- θ method and the Newmark- β method may be employed to solve Eq. (19). It should be noted, however, that the finite element model of the hydrofoil and the panel model of the fluid result in a huge system of linear algebraic equations. The required computation resource for solving Eq. (19) is considerable. The modal superposition method is often combined with the direct time-integration method to achieve fast computations in structural dynamics analyses. However, the added mass and damping matrices in Eq. (19) are asymmetric matrices, and this equation may not be calculated by the modal superposition method directly. A modal reduce technique is employed in the present analysis. The first N_{mode} wet mode vectors of the elastic hydrofoil can be combined to form a modal matrix, given as: Ψ_k . The displacement vector may be expressed in terms of the modal vectors ψ_k^i and generalized coordinate $u_k^i(t)$, given as: $q_k = \sum_{i=1}^{N_{\text{mode}}} u_k^i(t) \psi_k^i$. In doing so, Eq. (19) can be written as

$$\hat{M}_k \ddot{u}_k(t) + \hat{C}_k \dot{u}_k(t) + \hat{K}_k u_k(t) = \hat{F} \quad (22)$$

where

$$\Psi_k = \{\psi_k^1, \psi_k^2, \dots, \psi_k^{N_{\text{mode}}}\},$$

$$u_k(t) = [u_k^1(t), u_k^2(t), \dots, u_k^{N_{\text{mode}}}(t)]^T,$$

$$\hat{M}_k = \Psi_k^T [M + M_w(k)] \Psi_k,$$

$$\hat{C}_k = \Psi_k^T [C + C_w(k)] \Psi_k,$$

$$\hat{K}_k = \Psi_k^T K \Psi_k,$$

$$\hat{F} = \Psi_k^T \{F\}$$

Now, the total number of degrees of freedom involved in Eq. (22) is reduced to N_{mode} , and Eq. (22) can be solved very efficiently by using the Wilson- θ method. Once $u_k(t)$ is determined, q_k can be calculated by: $q_k = \sum_{i=1}^{N_{\text{mode}}} u_k^i(t) \psi_k^i$.

2. Validation

2.1 FEM validation

To confirm the validity of the present method for the structural modeling, numerical results of a cantilever NACA0015 hydrofoil obtained by present method are compared with those solutions of finite element analyses. The finite element solutions are obtained by using commercial software ANSYS. A total number of 24 600 eight-noded SOLID 45 elements are employed for the finite element modeling of the hydrofoil in ANSYS, see Fig. 2. The hydrofoil is of unit chord length and its span length is 4 m. The material properties of the hydrofoil are: density $\rho = 7800 \text{ kg/m}^3$, Poisson's ratio $\mu = 0.3$, and Young's modulus $E = 210 \text{ GPa}$. For free vibration analysis, the structural damping of the hydrofoil is neglected. In the present analysis, the finite element mesh of the hydrofoil surface coincides with that of the panel model, and a uniform mesh discretization is employed in the thickness direction of the hydrofoil. The natural frequencies of the first five natural frequencies obtained by the present method and those from the finite element analysis are listed in Table 1. In Table 1, N_s , N_c and N_t are the numbers of elements discretized in the span, chord and thickness directions of the hydrofoil, respectively. In order to show the convergent rate of the present method, different meshes are employed for the structural modeling of the hydrofoil. It is observed from Table 1 that the present results converge very rapidly, and the discrepancy between the present converged solutions and the finite element results is very small. For further validation of the present method, the forced vibration problem of the hydrofoil is examined. A point force

Table 1 Comparison of the first five natural frequencies and displacement amplitude of the hydrofoil in the z direction (in air)

$N_s \times N_c \times N_t$	Natural frequencies/Hz					Displacement /m
	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	
20×14×2	9.3677	42.1428	58.2913	63.7482	162.7358	0.4561×10^{-6}
40×20×3	7.4171	41.6570	46.0491	62.5962	127.3911	0.7537×10^{-6}
50×20×4	7.1313	41.5880	44.2556	62.4110	122.2874	0.8255×10^{-6}
60×14×2	7.0641	41.5460	43.8330	62.2493	121.0662	0.8463×10^{-6}
60×30×4	6.9817	41.5641	43.3022	62.3525	119.5341	0.8659×10^{-6}
FEM/BEM	6.6000	40.8400	41.5600	61.9000	112.6000	1.0000×10^{-6}

(see Fig. 2) is applied at one end point of the hydrofoil, which is expressed as: $f = \sin 6\pi t$. The amplitudes of the displacement measured at the driving force point in z direction are shown in Table 1. The present results agree well with those finite element solutions.

**Fig. 2** (Color online) Finite element model of the NACA 0015 hydrofoil

2.2 Fluid-structure interaction analysis

This section is concerned with the validation of the present method for the FSI analysis of hydrofoils. The accuracy of the obtained added mass and damping matrices by the present method is examined. In the present work, the validity of the added mass and damping matrices is confirmed indirectly by calculating the natural frequencies and dynamic responses of the hydrofoil in a stationary flow. The same hydrofoil studied in the Section 2.1 (see Fig. 2) is examined herein, except that the hydrofoil is immersed in a stationary flow. It can be inferred from

Appendix B that the vibration modes of a hydrofoil immersed in stationary flow based on the fluid-structure interaction analysis are the same as those based on the acoustic-solid interaction analysis if the speed of sound of the fluid is set to infinity. In addition, the structural responses calculated by the two types of analyses for the hydrofoil are the same. The density of the water is taken as: $\rho = 1000 \text{ kg/m}^3$. The present results for the FSI analysis of the hydrofoil are compared with those solutions obtained from the acoustic-structure interaction based on the finite element method and the boundary element method. The finite element model of the hydrofoil is constructed in ANSYS and the acoustic model of the fluid is formulated in Virtual.Lab Acoustics. The comparison of the mode frequencies and the displacement amplitudes are shown in Table 2. It is found that the present results are compared well with those solutions obtained from the coupled finite element/boundary element method. This validates the accuracy of the present solutions.

The convergence rate of the present method for predicting the unsteady forces in the z direction is investigated. The same hydrofoil examined in the previous section is considered, and different meshes are employed to compute the unsteady forces, as presented in Table 3. The inflow velocity is assumed as 10 m/s, and the inflow attack angle is taken as 10° . The driving frequency is specified as 14 Hz. In Table 3, $|F_z|$ represents the amplitude of the unsteady reactions in z direction and N_w represents the num-

Table 2 Comparison of the first five natural frequencies and displacement amplitude of the hydrofoil in the z direction (in water)

$N_s \times N_c \times N_t$	Natural frequencies/Hz					Displacement /m
	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	
20×14×2	6.8265	41.3151	41.7554	44.7063	114.9080	1.066×10^{-6}
40×20×3	5.4281	33.4094	41.2745	43.2951	91.4772	0.902×10^{-6}
50×20×4	5.2190	32.1373	41.2061	43.0594	87.9315	1.006×10^{-6}
60×14×2	5.1514	31.7138	41.1643	42.5214	86.7386	0.902×10^{-6}
60×30×4	5.1199	31.5224	41.1825	43.1947	86.1607	1.066×10^{-6}
FEM/BEM	4.8000	30.1000	38.9600	48.4500	84.9100	1.170×10^{-6}

bers of elements discretized along the wake direction. It is observed from Table 3 that the numerical solutions of the present method converge very rapidly as the number of the mesh elements and the vibration modes truncated in the superposition method are increased. The results show that solution size $50 \times 8 \times 2 \times 80 \times 100$ is sufficient to obtain reasonably accurate results, and this solution size will be employed in the following analysis.

Table 3 Convergence of the unsteady reactions in z direction with respect to the number of mesh elements and truncated vibration modes

$N_s \times N_c \times N_t \times N_w \times N_{\text{mode}}$	$ F_z /N$
$60 \times 16 \times 3 \times 80 \times 200$	1.28
$50 \times 8 \times 2 \times 80 \times 100$	1.18
$40 \times 6 \times 2 \times 80 \times 200$	1.33
$50 \times 8 \times 2 \times 80 \times 800$	1.23
$50 \times 8 \times 2 \times 200 \times 200$	1.25

3. Results and discussions

3.1 The added mass characteristics

The frequency-domain panel method in conjunction with the FEM has been employed to investigate the effect of the excitation frequency on the wet modes due to the added mass. Both symmetry and asymmetry NACA 0015 hydrofoils are considered, and the hydrofoils are clamped at their roots. The apse line equation for asymmetry NACA 0015 is given by: $y_c = 4x - 4x^2$, while the ratio of sagitta to chord length is taken as 0.1. The chord lengths for both hydrofoils are 1 m and the span lengths are chosen as 4 m. The material parameters of the hydrofoils are given as follows: Young's modulus $E = 210$ GPa, density $\rho_s = 7800$ kg/m³ and Poisson's ratio $\mu = 0.3$. The density of the fluid is $\rho = 1000$ kg/m³. The inflow velocity is assumed to be $V = 1.00$ m/s and attack angle α is 10° . The effect of the structural damping of the hydrofoil is not considered. The driving frequencies of the forces at the free end are specified as: $f = 0.46$ Hz, 2.30 Hz and 4.60 Hz for the symmetry hydrofoil, while for the asymmetrical hydrofoil, the driving frequencies are chosen as: $f = 0.40$ Hz, 2.00 Hz and 4.00 Hz. In the cases of the hydrofoil immersed in air and water, the first ten natural frequencies of the two hydrofoils predicted by the present method are shown in Fig. 3. To demonstrate the effect of imaginary parts of the added mass matrix on the dynamic behaviors of the hydrofoil, three numerical models are developed here, and they are labeled as model #1, #2 and #3. The added terms generated by the real part and the

imaginary part of the added mass matrix are included in model #1. In model #2, the real part of the added mass matrix is considered but the imaginary part is neglected. Static water is considered in model #3. It is observed from Fig. 3 that for both the symmetry and asymmetry hydrofoils, the effect of the imaginary part of the added mass matrix on the natural frequencies of the hydrofoils can be neglected. The term $\rho \mathbf{Z}^T \Delta s \mathcal{H}(\mathbf{U}) \mathbf{Z}$ is dominant for the effect of the added mass, moreover, the difference between results of the hydrofoils immersed in static and flowing water is attributed to the wake term in $\mathbf{U}$. Due to the existence of the FSI, the natural frequencies corresponding to the wet modes of the hydrofoils are significantly smaller than those of dry modes of the hydrofoils. This indicates that the wet modes must be employed for predicting the unsteady performance of an elastic hydrofoil, even though the deformation of the hydrofoil is not large. The decrease in the fundamental frequency of the hydrofoil may lead to resonance of the hydrofoil in lower frequency range.

The effect of flow velocity on the wet modes due to the added mass is further investigated. The same hydrofoils considered previously are examined in the following. The excitation frequencies considered here are 0.40 Hz (low frequency) and 4.00 Hz (high frequency) for the symmetry hydrofoil, while 0.46 Hz (low frequency) and 4.60 Hz (high frequency) are considered for the asymmetry hydrofoil. For the lower frequency case, four models considered previously are displayed, and inflow velocity V are specified as 0.02 m/s, 0.20 m/s and 2.00 m/s. The computed results for different models are presented in Fig. 4. It should be noted that only the cases of the model #1, and the hydrofoil immersed in static water and air are examined in the high frequency situation. The inflow velocity V is specified as 0.02 m/s, 2.00 m/s, 10.00 m/s, 50.00 m/s, 100.00 m/s, 200.00 m/s and 500.00 m/s. The computed results are shown in Fig. 5. The results in Figs. 4 and 5 show that for both the symmetry and asymmetry hydrofoils, the term $\rho \mathbf{Z}^T \Delta s \mathcal{H}(\mathbf{U}) \mathbf{Z}$ is a good approximation for predicting the wet modes of the hydrofoils in the case of hydrofoils immersed in static water. However, when the inflow velocity is relative large, the natural frequencies corresponding to the wet modes of the hydrofoils determined by $\rho \mathbf{Z}^T \Delta s \mathcal{H}(\mathbf{U}) \mathbf{Z}$ will be lower than those of the hydrofoils immersed in static water.

The effect of attack angle α on the wet modes due to the added mass is studied. The same hydrofoil is considered here with attack angle given as: $\alpha = 10^\circ$, $\alpha = 0^\circ$. The excitation frequencies considered here are 6.80 Hz for the symmetry hydrofoil and

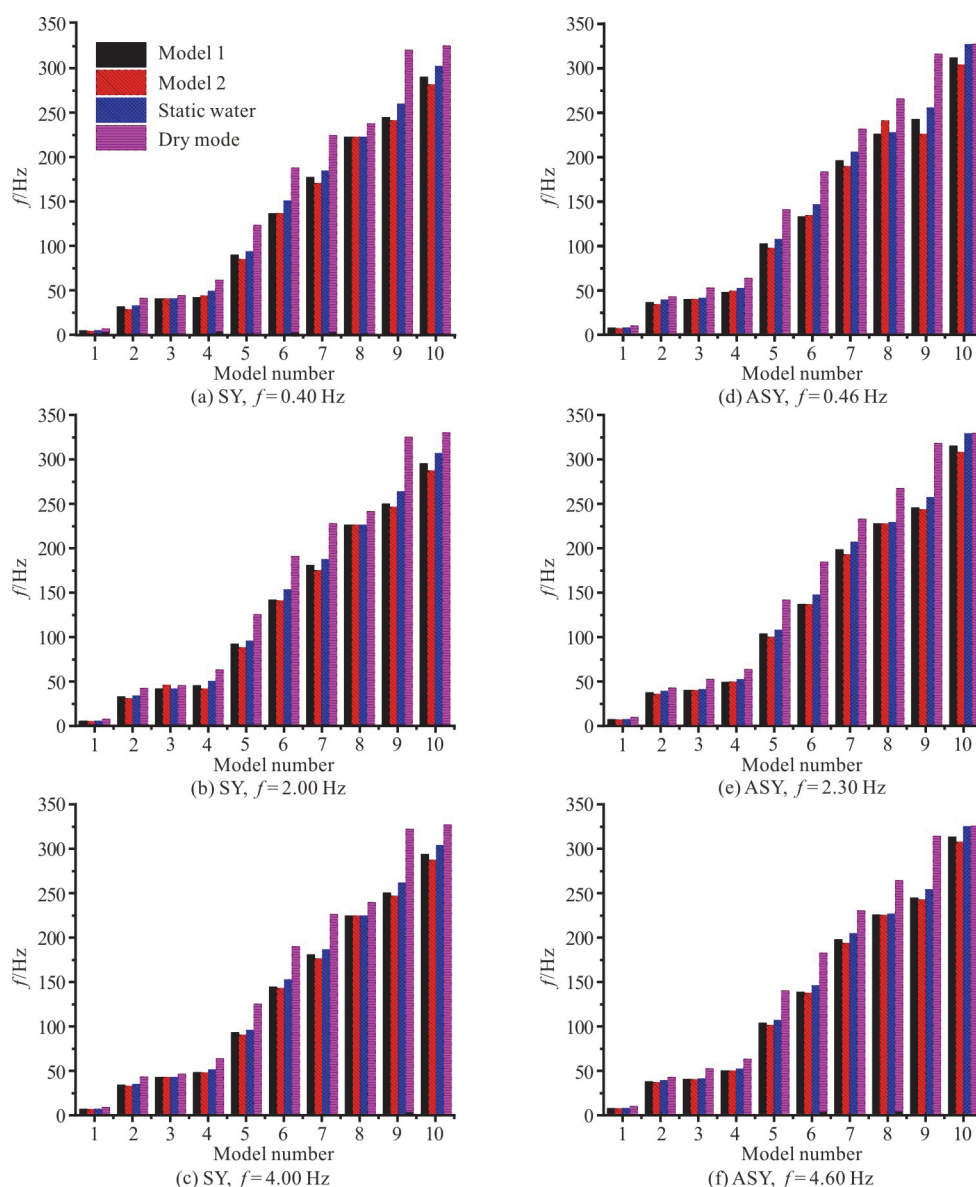


Fig. 3 (Color online) Comparison of the first ten natural frequencies of NACA 0015 hydrofoil

8.00 Hz for the asymmetry hydrofoil. Figure 6 shows the frequency ratios of the hydrofoil of attack angle $\alpha = 10^\circ$ to that of hydrofoil of $\alpha = 0^\circ$ for the first five modes. For the case of inviscid flow without cavitation, the attack angle has little effect on wet modes of the hydrofoil.

The effect of the elastic modulus on wet modes of the hydrofoil is further examined for the same hydrofoil considered previously. The modulus of the hydrofoil is taken as $E = 21$ GPa. The comparison of the first ten natural frequencies for the NACA 0015 under different inflow velocities is presented in Fig. 7. It is found that the natural frequencies of the hydrofoil are similar to those of the hydrofoil with elastic modulus $E = 210$ GPa. The decrease in the frequencies of the wet modes for the hydrofoil of elastic

modulus $E = 210$ GPa is of the same level of the dry mode frequencies. This can be explained by the linear theory with small deformation considered here.

3.2 The added damping characteristics

To demonstrate the effect of the added damping on the dynamic properties of the hydrofoils, the unsteady resultant forces of the hydrofoils measured in the z direction are calculated. The hydrofoils considered in previous section are examined herein.

The driving frequencies of the excitation forces are taken as 2.00 Hz (fundamental frequency of the hydrofoil immersed in air is 2.27 Hz for $E = 21$ GPa) and 6.8 Hz for the symmetry hydrofoil (fundamental frequency of the hydrofoil immersed in air is 7.20 Hz for $E = 210$ GPa), while 2.40 Hz (fundamental fre-

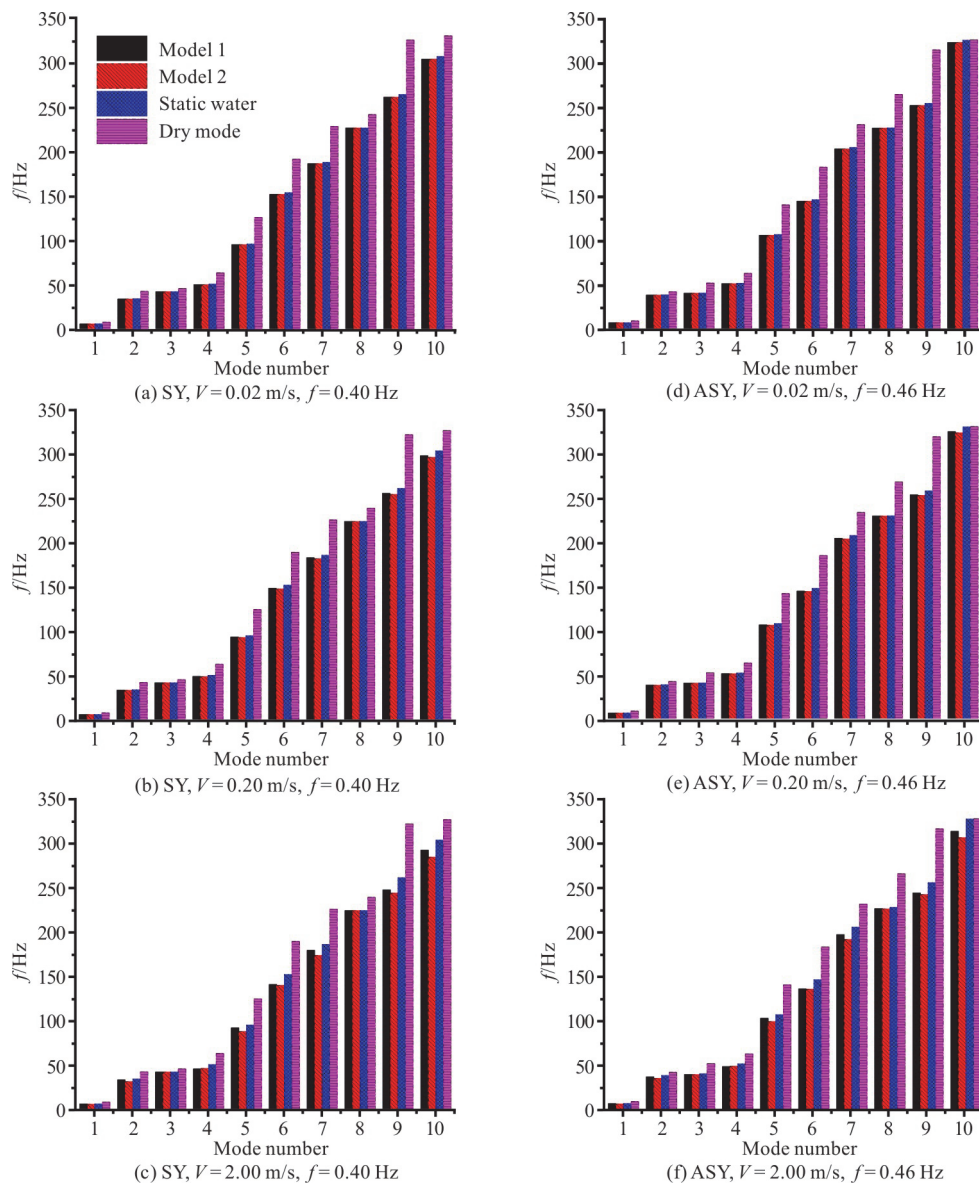


Fig. 4 (Color online) Comparison of the first ten natural frequencies of the NACA 0015 predicted by different models

quency of the hydrofoil in air is 2.60 Hz for $E = 21$ GPa and 8.00 Hz are considered for the asymmetry hydrofoil (fundamental frequency of the hydrofoil in air is 8.30 Hz for $E = 210$ GPa). However, in order to eliminate the influence of the added mass on the vibrations of the hydrofoils, only the added damping matrix is considered in Eq. (19). The two models in water examined in the previous section are considered here to analyze the terms generated due to the imaginary part of the added damping matrix. The comparisons of the unsteady force resultants of the hydrofoils measured in the z direction are presented in Fig. 8. Different inflow velocities are considered, and in all cases, the attack angle of the hydrofoils is taken as $\alpha = 10^\circ$. It is observed from Fig. 8 that the imaginary part of the

added damping matrix can be neglected for both the symmetry and asymmetry hydrofoils, when the inflow velocity is less than 0.02 m/s or larger than 50.00 m/s. As the inflow velocity is relatively large, the added damping will significantly affect the unsteady performance of the hydrofoils. This is due to the fact that the term $[V_0]_{x_i}$ exists in the added damping matrix. For the symmetry NACA 0015 hydrofoil, the amplitude of the unsteady resultant force is smaller than that of the driving force when the inflow velocity is taken as 500.00 m/s for hydrofoils with elastic modulus $E = 21$ GPa, $E = 210$ GPa. However, the amplitude of the unsteady resultant force of the asymmetry hydrofoil is larger than that of the driving force in all cases. This implies that the geometrical configuration of the hydrofoils will significantly affect

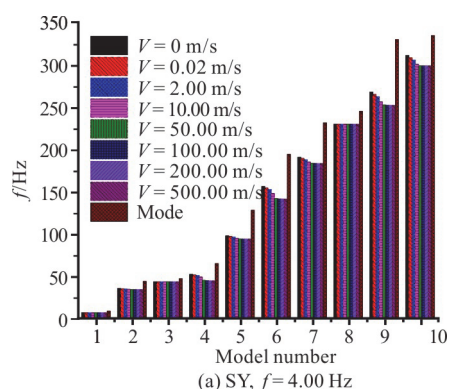
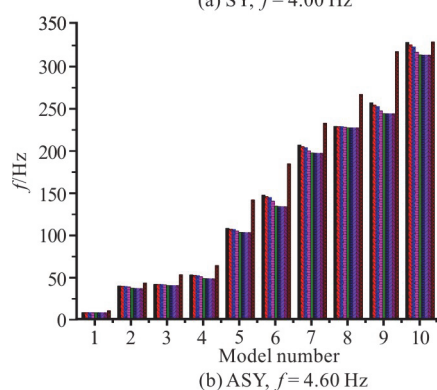
(a) SY, $f = 4.00$ Hz(b) ASY, $f = 4.60$ Hz

Fig. 5 (Color online) Comparison of the first ten natural frequencies for the NACA 0015 under different inflow velocities ($E = 210$ GPa)

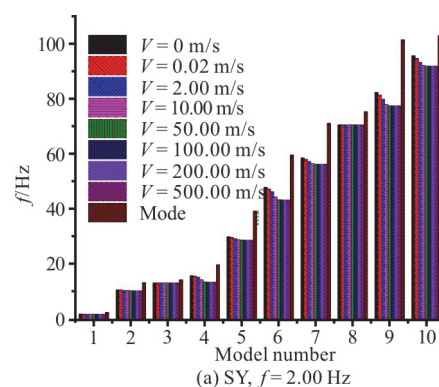
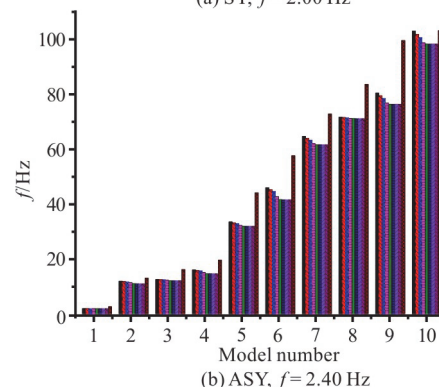
(a) SY, $f = 2.00$ Hz(b) ASY, $f = 2.40$ Hz

Fig. 7 (Color online) Comparison of the first ten natural frequencies for the NACA 0015 under different inflow velocities ($E = 21$ GPa)

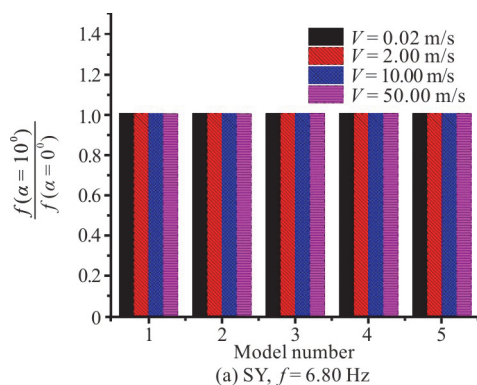
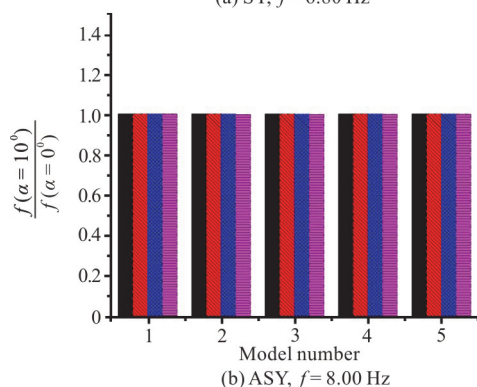
(a) SY, $f = 6.80$ Hz(b) ASY, $f = 8.00$ Hz

Fig. 6 (Color online) Comparison of the first five natural frequencies ratio by different attack angle for the NACA 0015 under different inflow velocities

their unsteady performance.

Finally, the effect of the attack angle α on the added damping is studied. The modulus of the hydrofoil is taken as $E = 210$ GPa, and the attack angle is given as $\alpha = 10^\circ$, $\alpha = 0^\circ$. The excitation frequencies considered here are 6.80 Hz for the symmetry hydrofoil and 8 Hz for the asymmetry hydrofoil. Figure 9 shows the ratio of the unsteady resultant forces for hydrofoil of attack angle $\alpha = 10^\circ$ to that of the hydrofoil of $\alpha = 0^\circ$. The results show that in the case of inviscid flow without cavitation, the attack angle has insignificant effect on added damping.

4. Conclusions

The FSI analysis of hydrofoils with small deformations is examined in the present work. A 3-D potential-based panel method is developed for the modeling of the fluid, and the 3-D finite element method is employed for the structural modeling of the hydrofoils. The frequency-dependent panel method is used to determine the added mass and damping matrices due to the FSI, while the finite element method is applied to evaluate the structural modes and resultant forces of the hydrofoils. A modal reduction

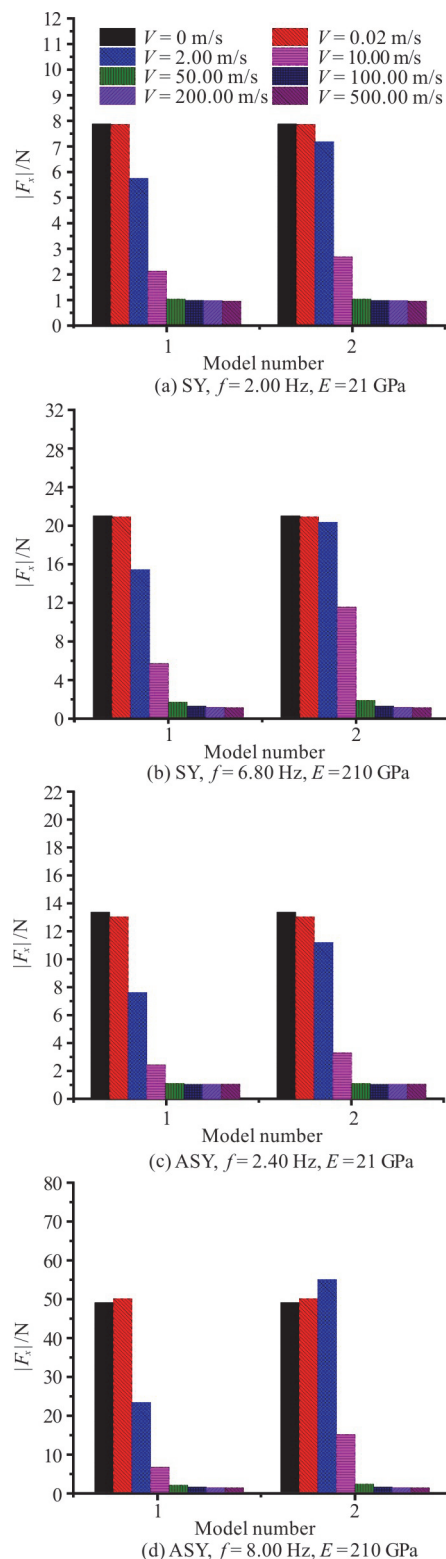


Fig. 8 (Color online) Comparison of the unsteady reactions in z direction for the NACA 0015 under different inflow velocities

technique combined with the Wilson- θ method is employed to overcome the low numerical efficiency

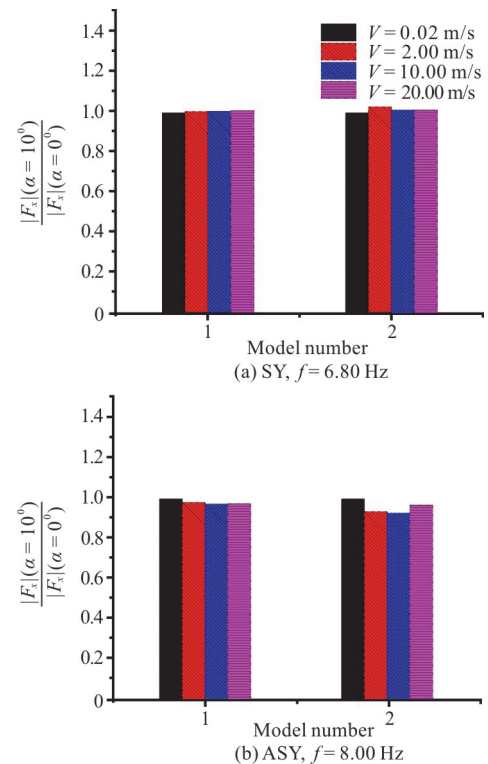


Fig. 9 (Color online) Comparison of the unsteady reactions ratio in z direction by different attack angle for the NACA 0015 under different inflow velocities

caused by the asymmetric added mass and damping matrices of the fluid. The validity of the present method is confirmed by comparing the results of the present method and those obtained from the coupled finite element/boundary element analyses. It is found that stationary flow is sufficient for analyzing the wet modes for both the symmetry and asymmetry hydrofoils. However, the imaginary part of the added damping matrix can be neglected for formulating the added damping matrix when the inflow velocity is either relatively large or small. Furthermore, in the case of large inflow velocity, the added damping may have significant effect on the prediction of the unsteady performance of the hydrofoil. The present method is restricted to FSI problems of hydrofoils involving inviscid flows without cavitation. It should be noted that the viscosity of the fluid may affect the added damping, and the phase change (i.e., cavitation) and multi-phase may influence the added mass and damping of hydrofoils. These complex effects are not considered in the present analysis. Considering the fluid viscosity in FSI problems requires more sophisticated numerical methods for the fluid, and in general, the boundary element methods are very difficult or even may not be suitable for solving such problems. In addition, for cavitation problems of flexible hydrofoils, the shapes of the cavitations are

unknown a priori, which should be solved iteratively. Time-dependent panel methods combined with the finite element methods are oriented to such problems.

Acknowledgement

This work was supported by the Scientific Research Foundation from Huazhong University of Science and Technology (Grant No. 2019kfyXJJS005).

References

- [1] Seeley C. E., Coutu A., Monette C. et al. Determination of hydrofoil damping due to fluid structure interaction using MFC actuators [C]. *54th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*, Boston, Massachusetts, USA, 2013.
- [2] Monette C., Nennemann B., Seeley C. et al. Hydrodynamic damping theory in flowing water [C]. *27th IAHR Symposium on Hydraulic Machinery and Systems*, Montréal, QC, Canada, 2014.
- [3] La Torre O. D., Escaler X., Egusquiza E. et al. Experimental investigation of added mass effects on a hydrofoil under cavitation conditions [J]. *Journal of Fluids and Structures*, 2013, 39: 173-187.
- [4] Kramer M. R., Liu Z., Young Y. L. Free vibration of cantilevered composite plates in air and in water [J]. *Composite Structures*, 2013, 95: 254-263.
- [5] Chae E. J., Akcabay D. T., Lelong A. et al. Numerical and experimental investigation of natural flow-induced vibrations of flexible hydrofoils numerical and experimental investigation of natural flow-induced vibrations of flexible hydrofoils [J]. *Physics of Fluids*, 2016, 28(7): 104-108.
- [6] Lelong A., Guiffant P., Andr J. An experimental analysis of the structural response of flexible lightweight hydrofoils in various flow conditions [C]. *International symposium on transport phenomena and dynamics of rotating machinery*, Honolulu, Hawaii, USA, 2016.
- [7] Liaghat T., Guibault F., Allenbach L. et al. Two-way fluid-structure coupling in vibration and damping analysis of an oscillating hydrofoil [C]. *ASME 2014 International Mechanical Engineering Congress and Exposition*, Montreal, Quebec, Canada, 2017.
- [8] Liu X., Zhou L., Escaler X. et al. Numerical simulation of added mass effects on a hydrofoil in cavitating flow using acoustic fluid-structure interaction [J]. *Journal of Fluids Engineering*, 2017, 139(4): 8.
- [9] Cao W., Xu H., Ren H. et al. Numerical study on characteristics of 3D cavitating hydrofoil [C]. *International Conference on Energy, Materials and Manufacturing Engineering*, Kuala Lumpur, Malaysia, 2015.
- [10] Wu Q., Huang B., Wang G. et al. Experimental and numerical investigation of hydroelastic response of a flexible hydrofoil in cavitating flow [J]. *International Journal of Multiphase Flow*, 2015, 74: 19-33.
- [11] Astolfi J. A., Lelong A., Bot P. Experimental analysis of hydroelastic response of flexible hydrofoils [C]. *5th High Performance Yacht Design Conference*, Auckland, New Zealand, 2015.
- [12] Hu S., Lu C., He Y. Fluid-structure interaction simulation of three-dimensional flexible hydrofoil in water tunnel [J]. *Applied Mathematics and Mechanics*, 2016, 37: 15-26.
- [13] Smith S. M., Venning J. A., Giosio D. R. et al. Cloud cavitation behavior on a hydrofoil due to fluid-structure interaction [J]. *Journal of Fluids Engineering*, 2019, 141(4): 041105.
- [14] Zeng Y. S., Yao Z. F., Zhou P. J. et al. Numerical investigation into the effect of the trailing edge shape on added mass and hydrodynamic damping for a hydrofoil [J]. *Journal of Fluids and Structures*, 2019, 88: 167-184.
- [15] Xie Z., Song P., Hao L. et al. Investigation on effects of fluid-structure-interaction (FSI) on the lubrication performances of water lubricated bearing in primary circuit loop system of nuclear power plant [J]. *Annals of Nuclear Energy*, 2020, 141: 107355.
- [16] Xie Z., Liu H. Experimental research on the interface lubrication regimes transition of water lubricated bearing [J]. *Mechanical Systems and Signal Processing*, 2020, 136: 106522.
- [17] Xie Z., Shen N., Ge J. et al. Analysis of the flow noises of the nuclear main pump caused by the high temperature liquid sodium in the two-circuit main loop liquid sodium pump system [J]. *Annals of Nuclear Energy*, 2020, 145: 107550.
- [18] Qu F., Chen J., Sun D. et al. A new all-speed flux scheme for the Euler equations [J]. *Computers and Mathematics with Applications*, 2019, 77(4): 1216-1231.
- [19] Li J., Rao Z., Su J. et al. A numerical method for predicting the hydroelastic response of marine propellers [J]. *Applied Ocean Research*, 2018, 74: 188-204.
- [20] Morino L., Kuo C. C. Subsonic potential aerodynamic for complex configurations: a general theory [J]. *AIAA Journal*, 1974, 12(2): 191-197.
- [21] Ghassemi H., Kohansal A. R. Numerical evaluation of various levels of singular integrals, arising in BEM and its application in hydrofoil analysis [J]. *Applied Mathematics and Computation*, 2009, 213(2): 277-289.
- [22] Tarafder M. S., Saha G. K., Mehedi S. T. Analysis of potential flow around 3-dimensional hydrofoils by combined source and dipole based panel method [J]. *Journal of Marine Science and Technology*, 2010, 18(3): 376-384.
- [23] Gao X. W. Evaluation of regular and singular domain integrals with boundary-only discretization-theory and Fortran code [J]. *Journal of Computational and Applied Mathematics*, 2005, 175(2): 265-290.
- [24] Hess J. L., Smith A. M. O. Calculation of potential flow about arbitrary bodies [J]. *Progress in Aeronautical Sciences*, 1967, 8: 1-138.
- [25] Hess J. L., Smith A. M. O. Calculation of non-lifting potential flow about arbitrary three-dimensional bodies [J]. *The Journal of Ship Research*, 1964, 8: 22-44.

Appendix A: Kinematic boundary conditions on hydrofoil surface

In Cartesian coordinate system as shown in Fig. 1, the material point $X = (x, y, z)$ on the hydrofoil surface has three displacement components, given as: $\delta^x(x, y, z, t)$, $\delta^y(x, y, z, t)$ and $\delta^z(x, y, z, t)$. Due to the vibration of the hydrofoil, the material point (x, y, z) moves to $(\bar{x}, \bar{y}, \bar{z})$, and $\bar{x}$, $\bar{y}$ and $\bar{z}$ are expressed as

$$[\bar{x}, \bar{y}, \bar{z}] = [x, y, z] + [\delta^x(x, y, z, t), \delta^y(x, y, z, t), \delta^z(x, y, z, t)] \quad (\text{A1})$$

Assume that $n(x, y, z) = 0$ is the geometrical function, which describes the undeformed surface, and the deformed surface can be described as

$$N(\bar{x}, \bar{y}, \bar{z}) = n[\bar{x} - \delta^x, \bar{y} - \delta^y, \bar{z} - \delta^z] = 0 \quad (\text{A2})$$

Supposing the normal vector of the deformed hydrofoil surface at point $(\bar{x}, \bar{y}, \bar{z})$ is $(N_{\bar{x}}, N_{\bar{y}}, N_{\bar{z}})$, one obtains:

$$\begin{aligned} N_{\bar{x}} &= n_x(1 - \delta_x^x x_{\bar{x}} - \delta_y^x y_{\bar{x}} - \delta_z^x z_{\bar{x}}) + \\ & n_y(-\delta_x^y x_{\bar{x}} - \delta_y^y y_{\bar{x}} - \delta_z^y z_{\bar{x}}) + \\ & n_z(-\delta_x^z x_{\bar{x}} - \delta_y^z y_{\bar{x}} - \delta_z^z z_{\bar{x}}), \\ N_{\bar{y}} &= n_x(-\delta_x^x x_{\bar{y}} - \delta_y^x y_{\bar{y}} - \delta_z^x z_{\bar{y}}) + \\ & n_y(1 - \delta_x^y x_{\bar{y}} - \delta_y^y y_{\bar{y}} - \delta_z^y z_{\bar{y}}) + \\ & n_z(-\delta_x^z x_{\bar{y}} - \delta_y^z y_{\bar{y}} - \delta_z^z z_{\bar{y}}), \\ N_{\bar{z}} &= n_x(-\delta_x^x x_{\bar{z}} - \delta_y^x y_{\bar{z}} - \delta_z^x z_{\bar{z}}) + \\ & n_y(-\delta_x^y x_{\bar{z}} - \delta_y^y y_{\bar{z}} - \delta_z^y z_{\bar{z}}) + \\ & n_z(1 - \delta_x^z x_{\bar{z}} - \delta_y^z y_{\bar{z}} - \delta_z^z z_{\bar{z}}) \end{aligned} \quad (\text{A3})$$

where δ_x^x represents the spatial derivative of δ^x with respect to x . $x_{\bar{x}}$ represents the spatial derivative of x with respect to $\bar{x}$. Other symbols, such as δ_y^y and $z_{\bar{y}}$ follow the similar definitions.

Taking the spatial derivative to both sides of Eq. (A1), one obtains

$$\begin{bmatrix} 1 + \delta_x^x & \delta_y^x & \delta_z^x \\ \delta_x^y & 1 + \delta_y^y & \delta_z^y \\ \delta_x^z & \delta_y^z & 1 + \delta_z^z \end{bmatrix} \begin{bmatrix} x_{\bar{x}} & x_{\bar{y}} & x_{\bar{z}} \\ y_{\bar{x}} & y_{\bar{y}} & y_{\bar{z}} \\ z_{\bar{x}} & z_{\bar{y}} & z_{\bar{z}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{A4})$$

Assuming the displacement of the hydrofoil is small, and substituting Eq. (A4) into Eq. (A3) gives

$$N_{\bar{x}} = n_x - n_x \delta_x^x - n_y \delta_x^y - n_z \delta_x^z,$$

$$N_{\bar{y}} = n_y - n_x \delta_y^x - n_y \delta_y^y - n_z \delta_y^z,$$

$$N_{\bar{z}} = n_z - n_x \delta_z^x - n_y \delta_z^y - n_z \delta_z^z \quad (\text{A5})$$

The normal velocity component of the fluid on the surface of the deformed hydrofoil must be zero. The following equation is obtained

$$\left[V_0(\bar{x}, \bar{y}, \bar{z}, t) + \nabla \phi(\bar{x}, \bar{y}, \bar{z}, t) - \frac{\partial \delta}{\partial t}(\bar{x}, \bar{y}, \bar{z}, t) \right] \cdot (N_{\bar{x}} \mathbf{i} + N_{\bar{y}} \mathbf{j} + N_{\bar{z}} \mathbf{k}) = 0 \quad (\text{A6})$$

where $V_0(\bar{x}, \bar{y}, \bar{z}, t)$ represents the velocity vector of uniform flow. $\nabla \phi(\bar{x}, \bar{y}, \bar{z}, t)$ denotes the disturbed velocity field due to the deformed hydrofoil. $(\partial \delta / \partial t)(\bar{x}, \bar{y}, \bar{z}, t)$ is the velocity vector of the point $(\bar{x}, \bar{y}, \bar{z})$.

Substituting Eq. (A5) and Taylor's expansion for $V_0(\bar{x}, \bar{y}, \bar{z}, t)$, $\nabla \phi(\bar{x}, \bar{y}, \bar{z}, t)$ and $(\partial \delta / \partial t)(\bar{x}, \bar{y}, \bar{z}, t)$ into Eq. (A6), the kinematic boundary condition imposed on the hydrofoil surface is obtained, given as

$$\frac{\partial \delta}{\partial t} \cdot \mathbf{n} = \frac{\partial \phi}{\partial \mathbf{n}} + V_0 \cdot \mathbf{n} + [(\delta \cdot \nabla) V_0 - (V_0 \cdot \nabla) \delta] \cdot \mathbf{n} \quad (\text{A7})$$

where $\mathbf{n}$ is the unit normal vector of the undeformed hydrofoil surface at point (x, y, z) , and $\mathbf{n}$ has the form: $(n_x, n_y, n_z) / \sqrt{n_x^2 + n_y^2 + n_z^2}$.

Since small elastic deformation of the hydrofoil is considered and V_0 will not change in space, one obtains:

$$\|\nabla \delta\| = o(\delta), \quad -\frac{\partial \phi}{\partial \mathbf{n}} \cong V_0 \cdot \mathbf{n} - \frac{\partial \delta}{\partial t} \cdot \mathbf{n} \quad (\text{A8})$$

The boundary conditions described in Eq. (A8) may be further decomposed into two different types of boundary conditions, given as:

$$-\frac{\partial \phi_s}{\partial \mathbf{n}} = V_0 \cdot \mathbf{n}, \quad \frac{\partial \phi_v}{\partial \mathbf{n}} = \frac{\partial \delta}{\partial t} \cdot \mathbf{n} \quad (\text{A9})$$

where Eq. (A9) is the kinematic boundary conditions for the rigid hydrofoil and elastic hydrofoil, respectively.

Appendix B: Comparison of fluid- and acoustic-structure interactions

In Cartesian coordinate system, the sound pressure field $\tilde{p}(\mathbf{R}, t)$ is governed by a 3-D wave equation. After assuming $\tilde{p}(\mathbf{R}, t) = p(\mathbf{R})e^{i\omega t}$, the Helmholtz equation can be obtained from the wave equation, written by

$$\nabla^2 p(\mathbf{R}) + k^2 p(\mathbf{R}) = 0 \quad (\text{B1})$$

where k is the wave number of the acoustic field, given as: $k = \omega/c_0$, and c_0 is the speed of sound.

In order to determine the sound pressure $p(\mathbf{R})$ in Eq. (B1), the Helmholtz equation is transformed into boundary integral equation, given as

$$2\pi p(\mathbf{R}_p) = \iint_{S_h} p(\mathbf{R}_Q) \frac{\partial}{\partial \mathbf{n}_Q} G(\mathbf{R}_p, \mathbf{R}_Q) dS - \iint_{S_h} G(\mathbf{R}_p, \mathbf{R}_Q) \frac{\partial}{\partial \mathbf{n}_Q} p(\mathbf{R}_Q) dS \quad (\text{B2})$$

The kinematic boundary conditions on the hydrofoil surface for the acoustic-structure coupling system require that normal velocity and acceleration on the hydrofoil surface are equal to those of fluid particle at the same location. After assuming $\tilde{\delta}_n(\mathbf{R}, t) = \delta_n(\mathbf{R})e^{i\omega t}$, then the kinematic boundary conditions can be written by

$$\frac{\partial p}{\partial \mathbf{n}} = \rho\omega^2 \delta_n \quad (\text{B3})$$

Substituting Eq. (B3) into Eq. (B2), then the nodal force can be written by

$$\mathbf{f} = \mathbf{Z}^T \Delta \mathbf{s} \{p\} = -\rho\omega^2 \mathbf{Z}^T \Delta \mathbf{s} \left[2\pi \mathbf{I} - \left(\frac{\partial G}{\partial \mathbf{n}} \right) \right]^{-1} \mathbf{GZq} \quad (\text{B4})$$

As Green's function can be expressed as

$$G(\mathbf{R}_p, \mathbf{R}_Q) = \frac{e^{-ik|\mathbf{R}_p - \mathbf{R}_Q|}}{|\mathbf{R}_p - \mathbf{R}_Q|} = \frac{\cos(k|\mathbf{R}_p - \mathbf{R}_Q|) - i\sin(k|\mathbf{R}_p - \mathbf{R}_Q|)}{|\mathbf{R}_p - \mathbf{R}_Q|} \quad (\text{B5})$$

The nodal force can be rewritten as

$$\mathbf{f} = -(\mathbf{M}_a \mathbf{q} + i\mathbf{C}_a \mathbf{q}) \quad (\text{B6})$$

where $\mathbf{M}_a$ and $\mathbf{C}_a$ are the added mass and added damping matrices due to the coupling of the acoustic and structural fields.

After a rigorous derivation, the following statement can be proven: only if $k=0$, $\mathbf{M}_a = \mathbf{M}_w|_{V_0=0}$, $\mathbf{C}_a = \mathbf{C}_w|_{V_0=0} = \mathbf{0}$.

In summary, the add-mass matrix due to the FSI in the stationary flow is equal to that of the acoustic-solid interaction when wave number of sound is zero. However, the added-damping matrices vanish in the case of $k=0$. Consequently, the governing equations for the FSI problem have the same solutions as those of the acoustic-structure interaction problem.